



Food and Agriculture
Organization of the
United Nations



Technical workshop for GCF project: Climate change and climate change impacts on São Tomé and Príncipe with a focus on fishery and tourism sectors in a Blue Economy context

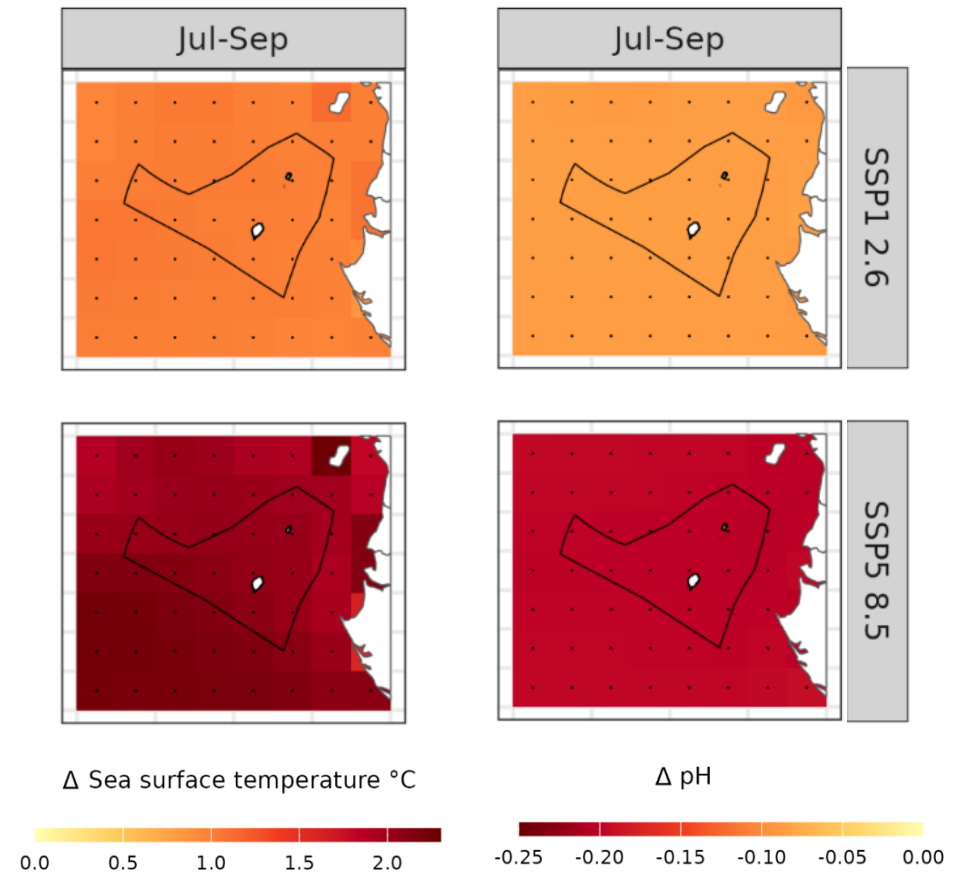
Sao Tome and Principe, November/December 2025

SPECIES DISTRIBUTION MODELLING USING MAXENT

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Fisheries in Sao Tome and Principe: the challenge

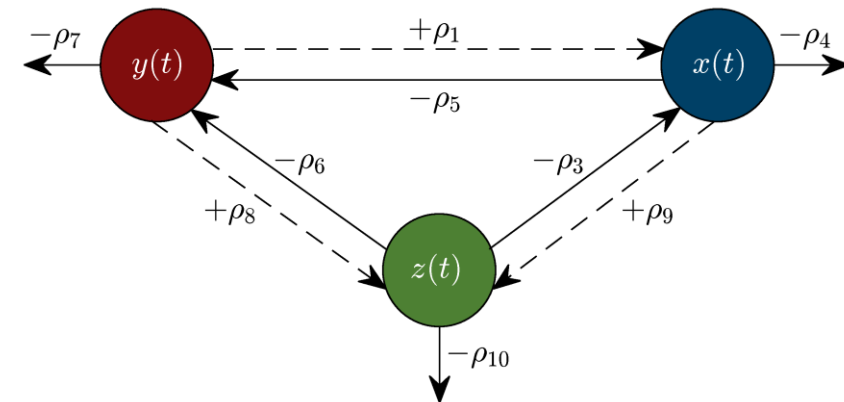
- Climate models project significant future oceanographic change
- Climate change is expected to affect fisheries by altering:
 - Fish reproduction
 - Fish densities
 - Fish migration patterns



Values are relative to
historical run

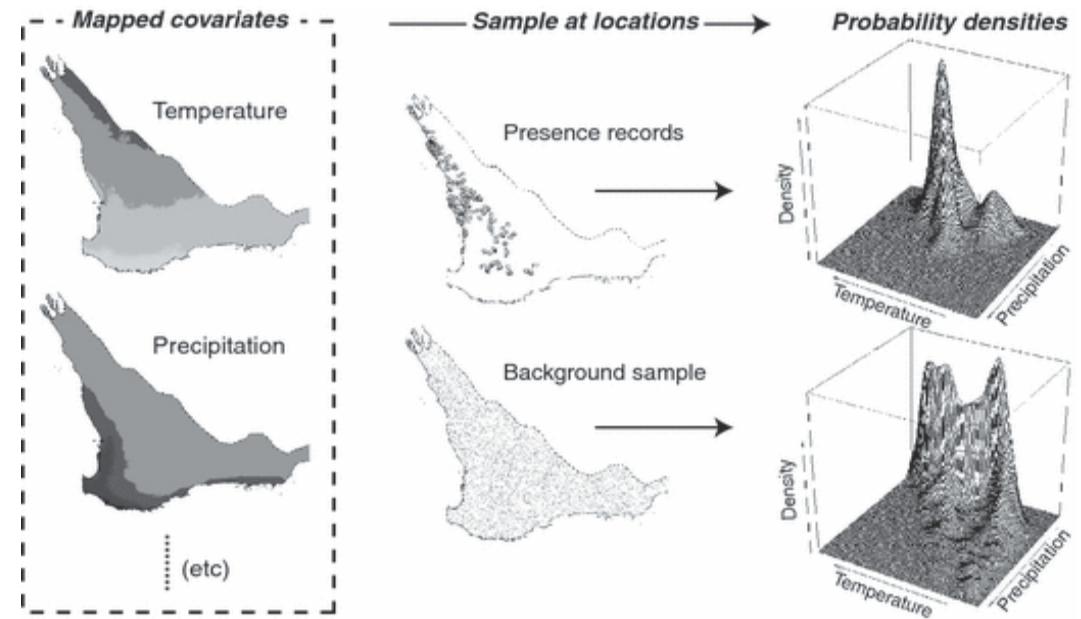
Species distribution modelling approaches

- Species distribution modelling can be used to assess climate impact on fish distributions
- **Most models are “empirical”** – find statistical associations between species occurrence and environmental parameters
 - MaxEnt
 - R-INLA
 - SdmTMB
- **A small number are “mechanistic”** – use equations that model how fish systems function and interact with climate



Empirical modelling – MaxEnt for species distributions

- For the STP report, we used an empirical modelling approach called “MaxEnt” or Maximum Entropy
 - Species presence records
 - Climatic values across a region
 - Compares probability distribution of associated climatic values against background data.
- Once model is trained, you can apply new climate projection data.



Occurrence probabilities calculated for:
Time periods: historical run, 2020-2049, 2050-2079
Climate scenarios: SSP1 2.6 & SSP5 8.5
Climate model: CMIP6 ensemble

Important considerations

Advantages of MaxEnt:

- **Handles presence-only data:** MaxEnt is well-suited for situations where only presence data is available, which is common in ecological research.
- **Handles complex relationships:** It can model complex interactions between environmental variables and species distributions.
- **Open-source software:** MaxEnt is freely available and open-source.
- **Widely used:** It's one of the most widely used SDM methods in ecology.

Potential Limitations:

- **Bias in background data:**

The choice of background data can influence the model results.

- **Assumptions about environmental relationships:**

MaxEnt assumes that environmental variables are the primary drivers of species distributions, which may not always be true.

- **Overfitting:**

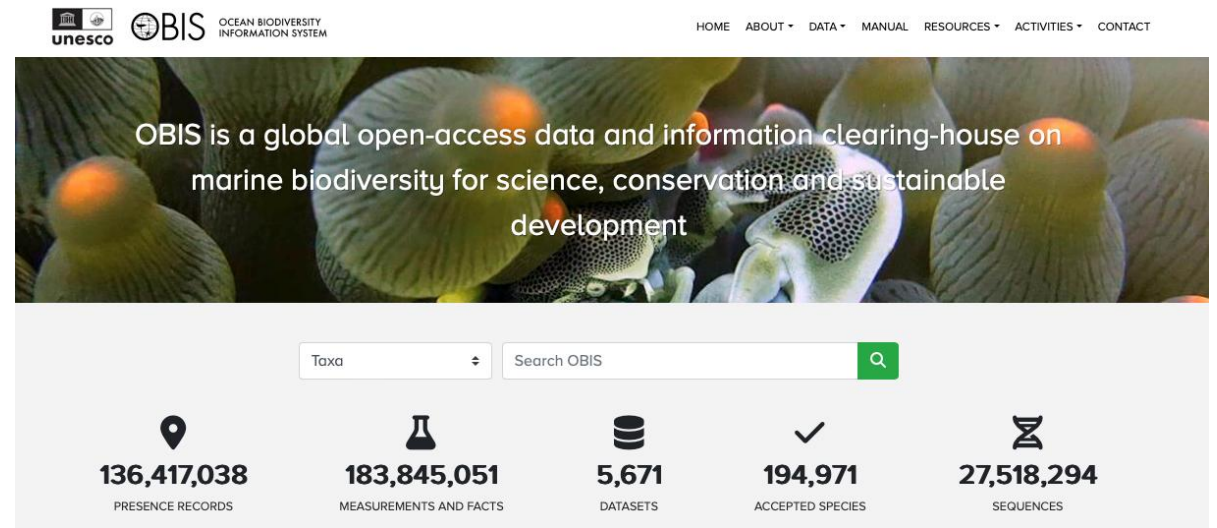
Overfitting can occur if the model is too complex for the available data, leading to poor predictive performance in new areas.

Data collection – Species presence data

- MaxEnt only requires presence data, not presence/absence data
- Data was acquired from the Ocean Biodiversity Information System (OBIS)

Required information:

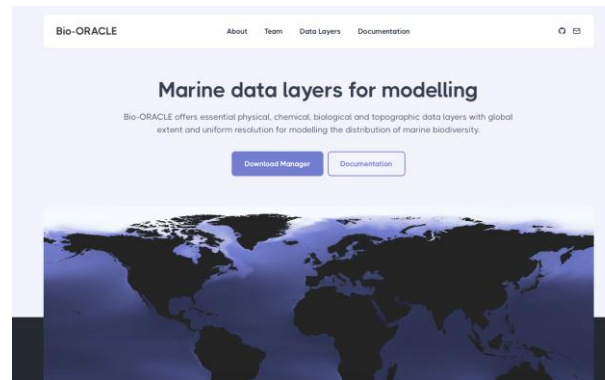
- Species name
- Historical Lon/Lat positions



Data collection – marine climatic data

- Historical and future climate data was acquired from Bio-Oracle:

- Ocean temperature
- Salinity
- Dissolved molecular oxygen
- pH



☒ Benthic layers [conditions along the sea bottom]

☒ Minimum depth ☐ Average depth ☐ Maximum depth

The benthic layers were produced with an interpolation process considering the geographic position and depth of cells, as inferred from a bathymetry layer. Since focal cells included a wide range of depth values, benthic layers are available for the minimum, average and maximum depths.

List of layers

Variable	Unit	Max	Mean	Min	Lt. Max	Lt. Min	Range
Ocean temperature	oC	☐	☐	☐	☐	☐	☐
Salinity	-	☐	☐	☐	☐	☐	☐
Sea water velocity	m.s-1	☐	☐	☐	☐	☐	☐
Sea water direction	degree	☐	☐	☐	☐	☐	☐
Nitrate	mmol . m-3	☐	☐	☐	☐	☐	☐
Phosphate	mmol . m-3	☐	☐	☐	☐	☐	☐
Silicate	mmol . m-3	☐	☐	☐	☐	☐	☐
Dissolved molecular oxygen	mmol . m-3	☐	☐	☐	☐	☐	☐
Iron	mmol . m-3	☐	☐	☐	☐	☐	☐
Primary productivity	mmol . m-3	☐	☐	☐	☐	☐	☐
pH	-	☐	☐	☐	☐	☐	☐
Bathymetry	m	☐	☐	☐	☐	☐	☐
Topographic slope	-	☐	☐	☐	☐	☐	☐
Topographic aspect	-	☐	☐	☐	☐	☐	☐
Topographic position index	-	☐	☐	☐	☐	☐	☐
Terrain ruggedness index	-	☐	☐	☐	☐	☐	☐

- Mean and range* values selected

☒ Future conditions (select) ▼

☐ SSP1-1.9 ☐ SSP1-2.6 ☐ SSP2-4.5 ☐ SSP3-7.0 ☐ SSP4-6.0 ☒ SSP5-8.5

Data layers reflecting future conditions were developed under the Shared Socioeconomic Pathway (SSP) scenarios of future climate change. These span from a "sustainability" scenario SSP1-1.9 following the target of Paris Agreement of reduced greenhouse gas emissions, to the "fossil-fuelled development" SSP5-8.5 scenario of high emissions and low challenges to adaptation.



Example: *Caranx crysos*

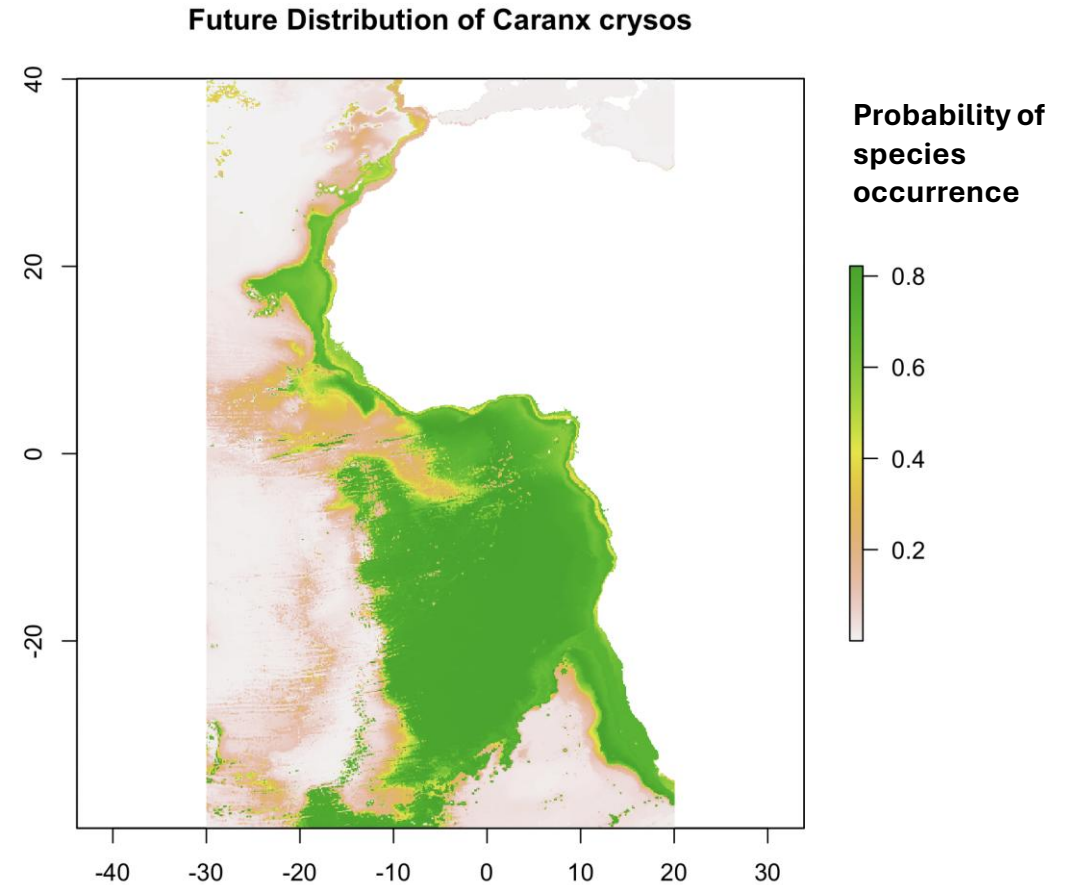
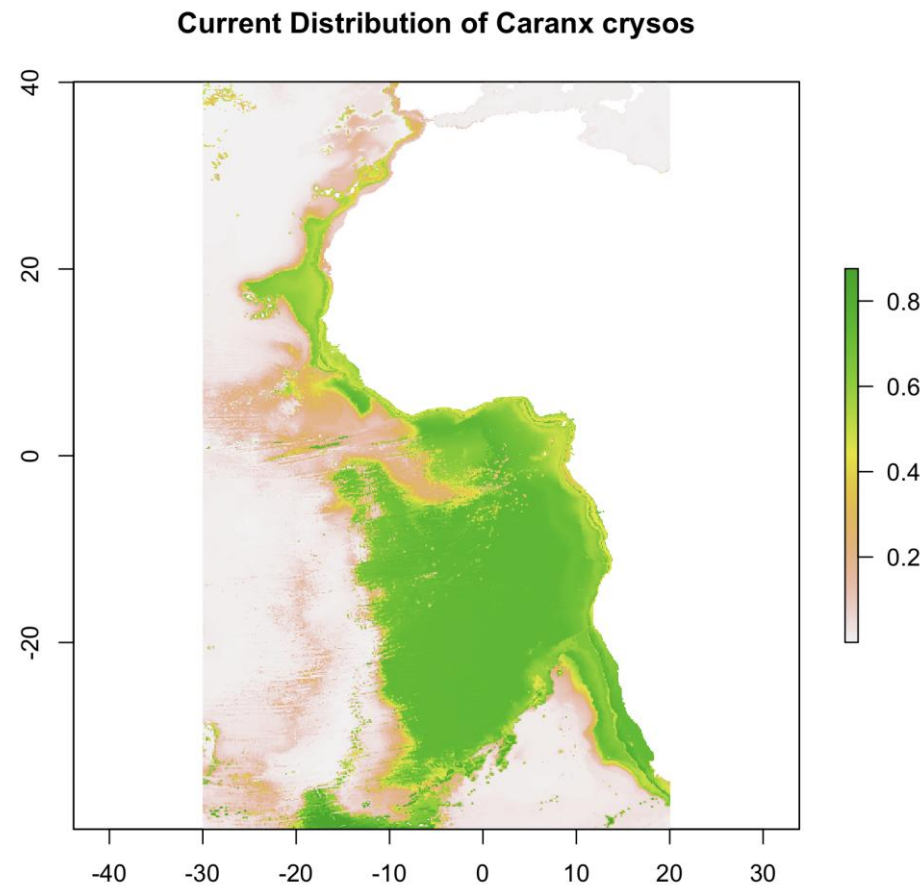
Empirical modelling – MaxEnt for species distributions

.2 = Range
.1 = Mean

- For *Caranx crysos*:
- Most important indicators:
 1. Dissolved molecular oxygen range
 2. Average ocean temperature

Variable	Percent contribution	Permutation importance
Average.DissolvedMolecularOxygen.2	52.4	34.9
Average.OceanTemperature.1	30.5	37.9
Average.Salinity.1	8.9	4.7
Average.Chlorophyll.2	3.2	6.9
Average.pH.1	2.5	3.3
Average.OceanTemperature.2	2.2	6.8
Average.Chlorophyll.1	0.3	5.6
Average.DissolvedMolecularOxygen.1	0	0
Average.pH.2	0	0
Average.Salinity.2	0	0

Empirical modelling – MaxEnt for species distributions



Empirical modelling – MaxEnt for species distributions

